

E5: Capstone Assignment

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Introduction

I initially chose the Galapagos Archipelago for this assignment as it is an area I have always been interested in, but in doing some research on the area I realised that it is actually very interesting in an oceanographic sense when focused on the interactions that occur between the bathymetry and chlorophyll-a.

It is made up of a series of island of the coast of Ecuador, sitting in the intersection of major ocean currents (De Falco et al., 2022). Being situated at the equator, its islands are dynamically influenced by the eastward-flowing Cromwell Current which is also known as the Equatorial Undercurrent (EUC). When this deep and nutrient-rich subsurface current collides with the steep volcanic platforms of the western island, particularly Isabela and Fernandina, it is forced upward (Karnauskas and Giglio, 2022).

The localised topographic upwelling injects high concentrations of essential macronutrients and dissolved iron into the sunlit photic zone, this fuels massive blooms of phytoplankton (Behrenfeld and Boss, 2014). This is a prime example of the Island Mass Effect (IME), where current-island interactions significantly amplify the chlorophyll-a concentrations relative to the surrounding oligotrophic waters (De Falco et al., 2022).

The broader Eastern Tropical Pacific is generally characterised by lower nutrient availability, the western side of the Galapagos Archipelago contrasts this as it is a persistent and highly localised biological hotspot. Choosing this region for an analysis of Chlorophyll-a allows for an investigation into how bathymetry interacts with physical oceanography to drive spatial and temporal variations in marine ecosystems.

Description of the datasets

Bathymetry

– Dimensions:	(lat: 2734, lon: 2276)		
▼ Coordinates:			
lat	(lat)	float64	-1.502 -1.501 ... 1.501 1.502
lon	(lon)	float64	-92.5 -92.5 -92.5 ... -90.0 -90.0
▼ Data variables:			
altitude	(lat, lon)	float64	...
▼ Attributes:			
title :	GMRT Grid		
history :	Projection: Cylindrical Equidistant		
	Extracted from the Global Multi-Resolution Topography (GMRT) Synthesis v4		
	https://www.gmrt.org		
	Created with GMRT GridServer		
	https://www.gmrt.org/services/gridserverinfo.php		
Conventions :	COARDS,CF-1.6		
GMT_version :	4.5.7		

Chlorophyll-a

► Dimensions:	(time: 12, lat: 4320, lon: 8640)		
▼ Coordinates:			
time	(time)	datetime64[ns]	1998-01-01 ... 1997-12-01  
lat	(lat)	float64	89.98 89.94 89.9 ... -89.94 -89.98  
lon	(lon)	float64	-180.0 -179.9 ... 179.9 180.0  
▼ Data variables:			
crs	()	int32	...  
chlor_a	(time, lat, lon)	float32	...  
▼ Attributes:			
CDI :	Climate Data Interface version ?? (http://mpimet.mpg.de/cdi)		
history :	Tue Apr 27 20:27:21 2021: cdo selvar, chlor_a ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_CHL-fv5.0.nc Tue Apr 27 19:42:57 2021: ncrat ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-01-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-02-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-03-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-04-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-05-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-06-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-07-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-08-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-09-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-10-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-11-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-12-fv5.0.nc ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_OCx_QAA-fv5.0.nc		
source :	rs_bbb_rmsd, chlor_a_log10_rmsd with --root-mean-square, and - with --total NASA SeaWiFS L1A and L2 R2018.0 LAC and GAC, MODIS-Aqua L1A and L2 R2018.0, MERIS L1B 3rd reprocessing inc OCL corrections, NASA VIIRS L1A and L2 R2018.0, OLCI L1B		
institution :	Plymouth Marine Laboratory		
Conventions :	CF-1.7		
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comment :	See summary attribute		
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creator_name :	Plymouth Marine Laboratory		
creator_url :	http://esa-oceancolour-cci.org		
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keywords_vocab...	none		
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number_of_optic... 14
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 project : Climate Change Initiative - European Space Agency
 references : <http://www.esa-oceancolour-cci.org/>
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 title : ESA CCI Ocean Colour Product
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 number_of_files_... 23
 summary : Data products generated by the Ocean Colour component of the European Space Agency Climate Change Initiative project. These files are monthly composites of merged sensor (MERIS, MODIS Aqua, SeaWiFS LAC & GAC, VIIRS, OLCI) products. MODIS Aqua and SeaWiFS were band-shifted and bias-corrected to MERIS bands and values using a temporally and spatially varying scheme based on the overlap years of 2003-2007. VIIRS was band-shifted and bias-corrected in a second stage against the MODIS Rrs that had already been corrected to MERIS levels, for the overlap period 2012-2013; and at the third stage OLCI was bias corrected against already corrected MODIS, for overlap period 2016-07-01 to 2019-06-30. VIIRS, MODIS, SeaWiFS and MERIS Rrs were derived from a combination of NASA's l2gen (for basic sensor geometry corrections, etc) and HYGEO Polymer v4.12 (for atmospheric correction). OLCI Rrs were sourced at L1b (already geometrically corrected) and processed with polymer. The Rrs were binned to a sinusoidal 4km level-3 grid, and later to 4km geographic projection, by Brockmann Consult's SNAP. Derived products were generally computed with the standard algorithm through SeaDAS. QAA IOPs were derived using the standard SeaDAS algorithm but with a modified backscattering table to match that used in the bandshifting. The final chlorophyll is a combination of OCI, OCI2, OC2 and OCx, depending on the water class memberships. Uncertainty estimates were added using the fuzzy water classifier and uncertainty estimation algorithm of Tim Moore as documented in Jackson et al (2017). and updated according to Jackson et al. (in prep).

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 CDO : Climate Data Operators version 1.9.3 (<http://mpimet.mpg.de/cdo>)

Figures

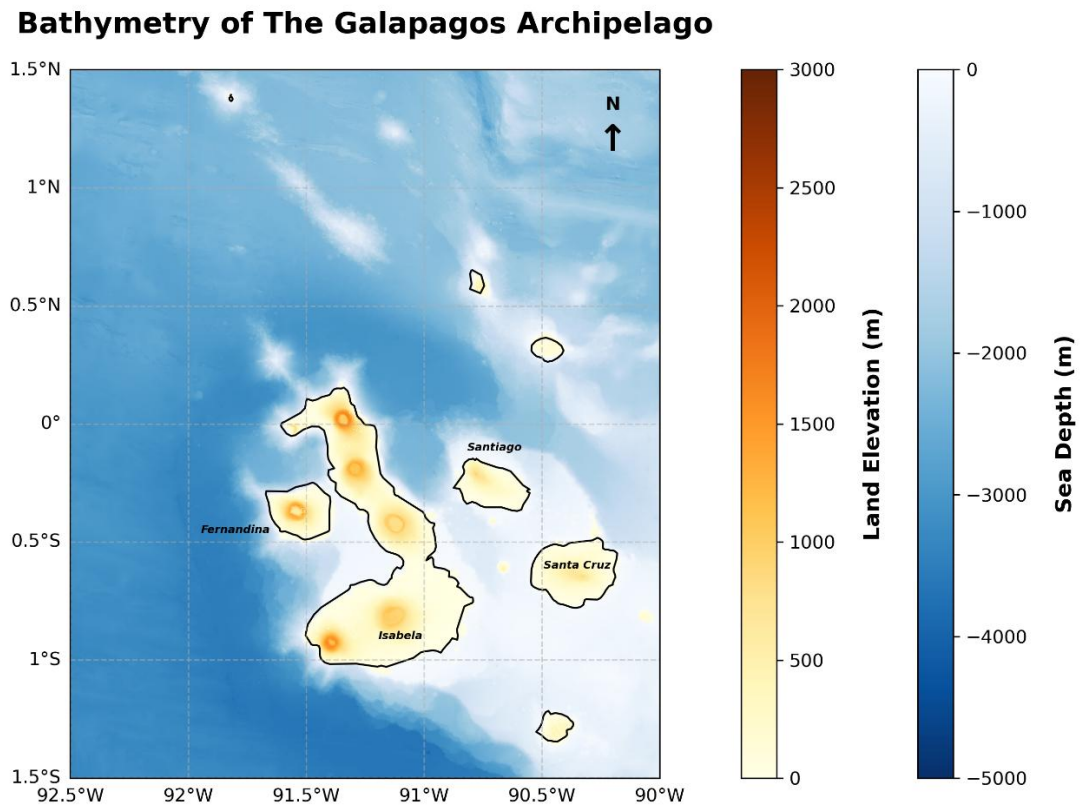


Figure 1: Bathymetric and topographic map of the Galapagos Archipelago, displaying the land elevation and sea depth in meters (m).

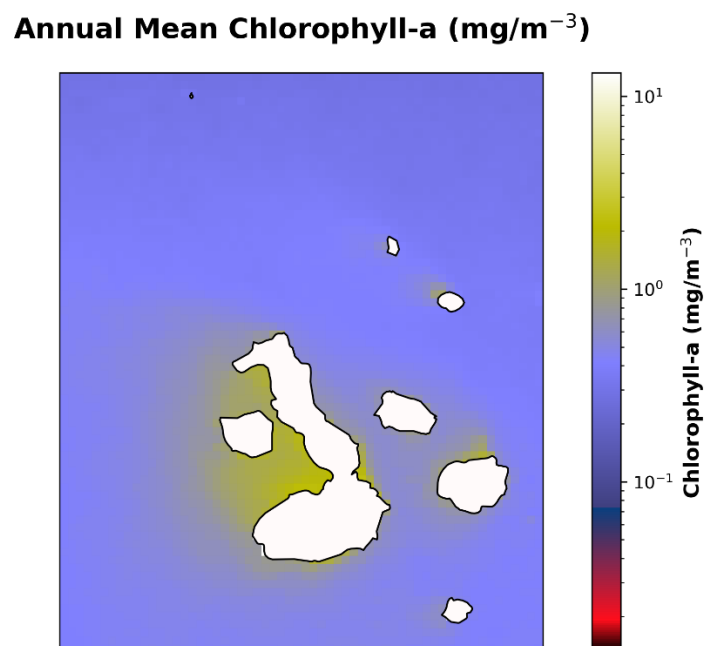


Figure 2: Spatial distribution of the annual mean Chlorophyll-a concentration surrounding the Galapagos Archipelago.

Monthly Chlorophyll-a concentrations of the Galapagos Archipelago

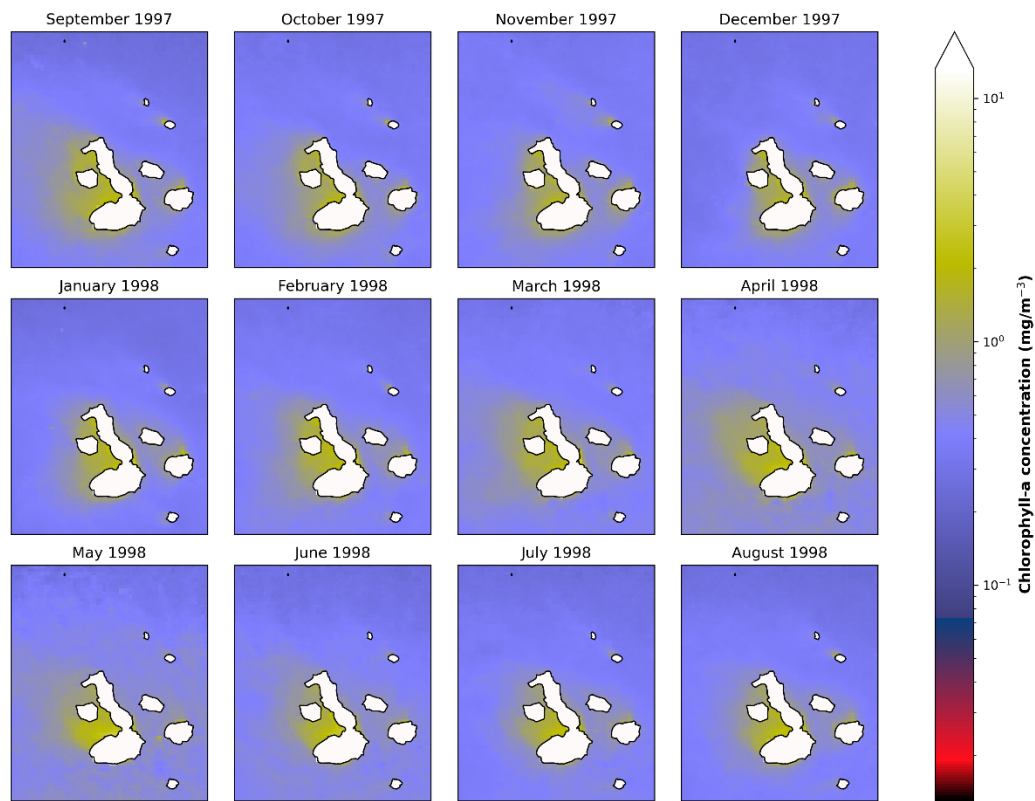


Figure 3: A time-series grid mapping the monthly variations in Chlorophyll-a concentrations around the Galapagos Archipelago for a year, starting September 1997 and ending in August 1998.

Seasonal Chlorophyll-a Comparison: Regional Average vs Local Hotspot

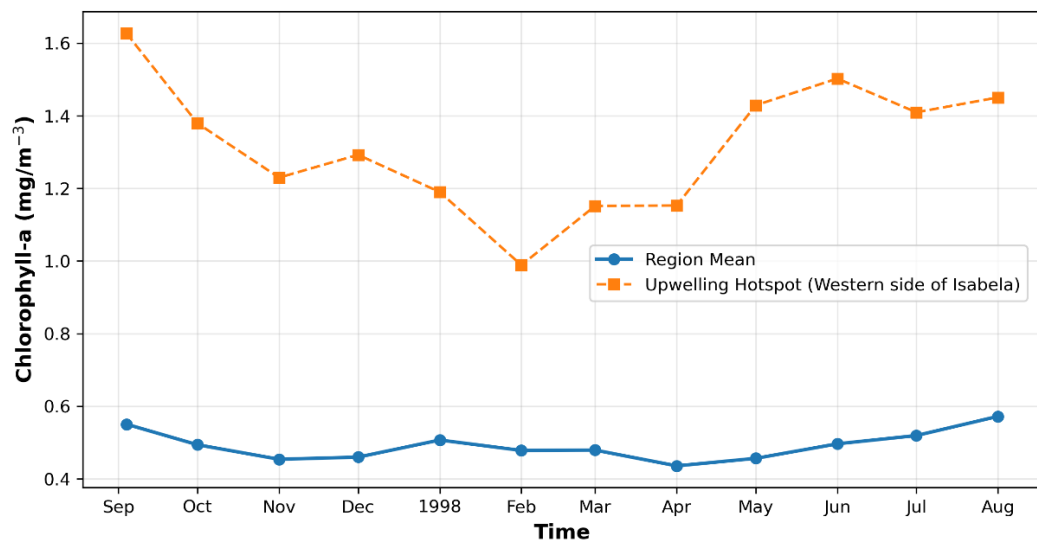


Figure 4: A line graph comparing the seasonal Chlorophyll-a concentrations from September 1997 to August 1998, contrasting the overall regional mean to the localized upwelling hotspot located on the western side of Isabela Island.

Disclaimer for use of AI:

I did make use of AI during the completion of this assignment. I mainly used AI when I ran into errors in the code that I was not able to fix myself. I have included some of the prompts I used below:

- “I have the following error ... how should I correct it?”
- “I would like to split the colorbar into 2 to display the land elevation and sea depth, how would I do that?”
- “I keep getting a syntax error, but I am unable to find the problem, how should I correct the code below to let it run?”
- “I am not getting the output I want, the coastlines and land are not appearing, but I have added in the code, and it runs without any errors. What is the problem?”

References

Behrenfeld, M.J. and Boss, E.S., 2014. Resurrecting the ecological underpinnings of ocean plankton blooms. *Annual Review of Marine Science*, 6(1), pp.167–194. Available at: <https://doi.org/10.1146/annurev-marine-052913-021325>.

De Falco, C., Desbiolles, F., Bracco, A. and Pasquero, C., 2022. Island Mass Effect: A review of oceanic physical processes. *Frontiers in Marine Science*, 9, p.894860. Available at: <https://doi.org/10.3389/fmars.2022.894860>.

Karnauskas, K.B. and Giglio, D., 2022. Argo reveals the scales and provenance of equatorial island upwelling systems. *Geophysical Research Letters*, 49(16), p.e2022GL098744. Available at: <https://doi.org/10.1029/2022gl098744>.